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Securing Microservice Architectures with Traefik and CrowdSec on openSUSE Server

Andi Sugandi

Yogyakarta | October, 2026

Talk record » https://youtu.be/MHeoOSrwBt8

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Andi Sugandi



Peminat FOSS and openSUSE Member Indonesia (sejak 2007)

- Mengajar
- Tim Infrastruktur TIK

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Securing Microservice Architectures with Traefik and CrowdSec on openSUSE Server

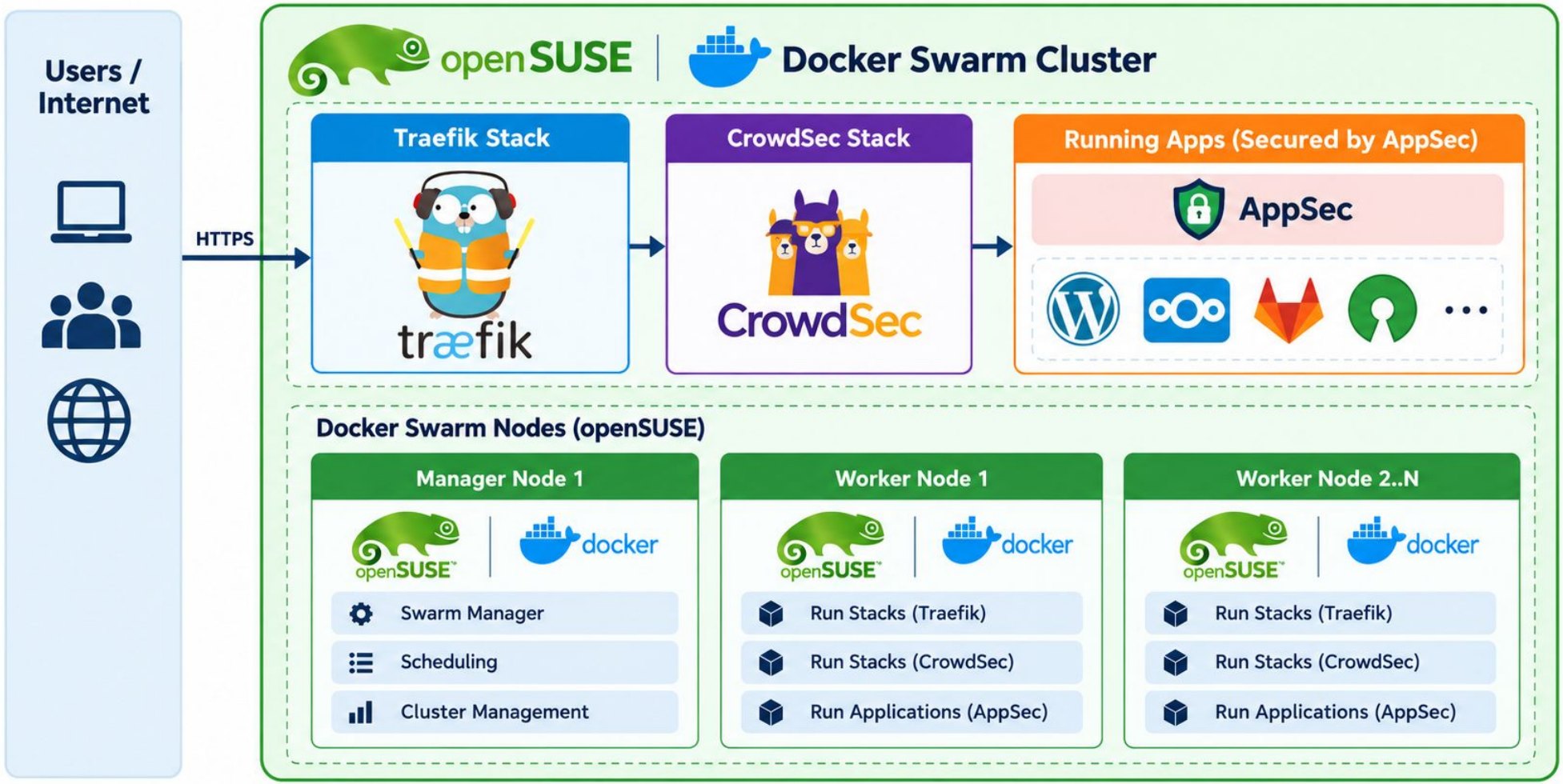
Andi Sugandi

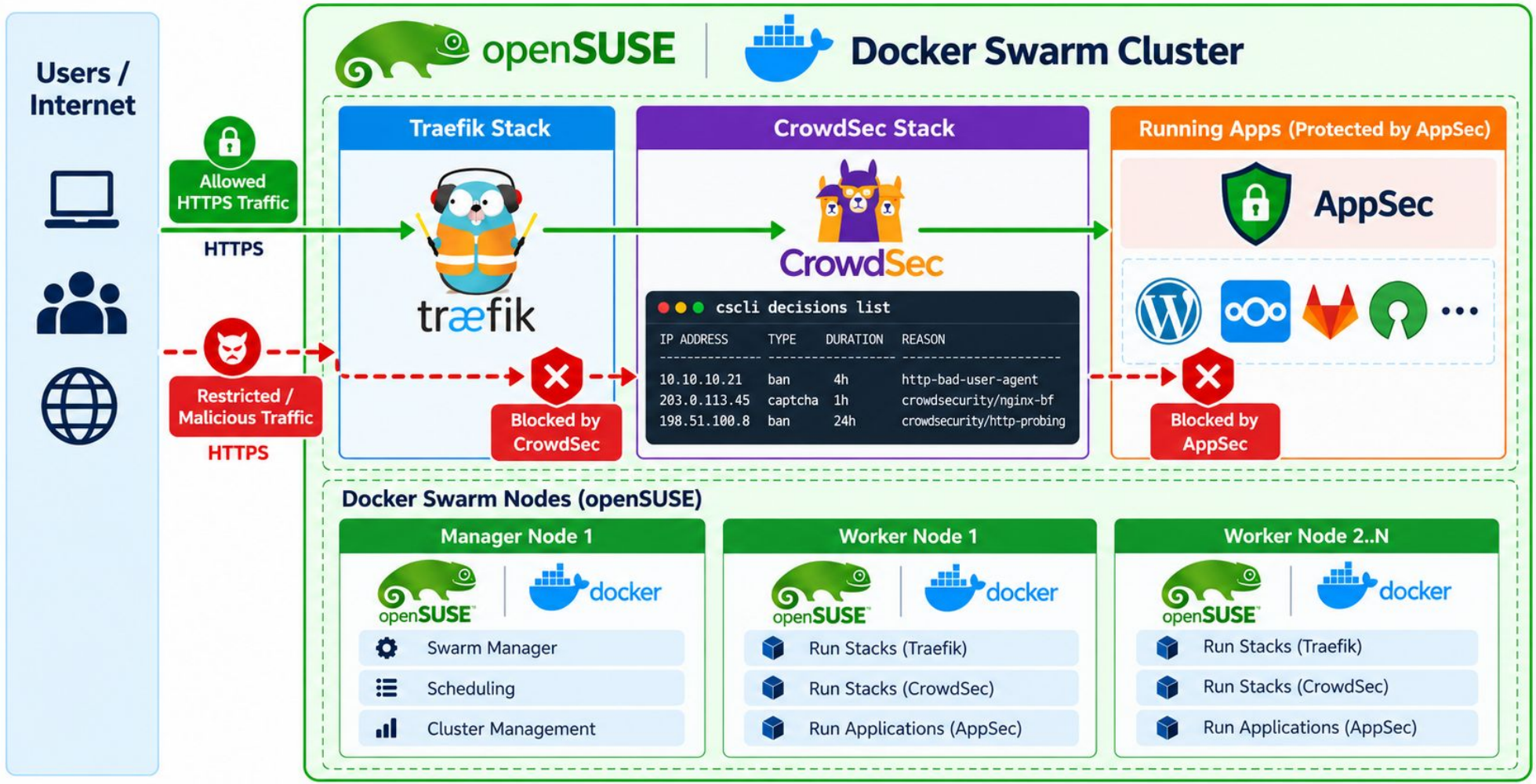
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Building a Resilient and Threat-Aware Reverse Proxy Perimeter


Users / Internet
HTTPS
openSUSE
Docker Swarm Cluster
Traefik Stack
traefik
CrowdSec Stack
CrowdSec
Running Apps (Secured by AppSec)
AppSec
Docker Swarm Nodes (openSUSE)
Manager Node 1
openSUSE
docker
Swarm Manager
Scheduling
Cluster Management
Worker Node 1
openSUSE
docker
Run Stacks (Traefik)
Run Stacks (CrowdSec)
Run Applications (AppSec)
Worker Node 2..N
openSUSE
docker
Run Stacks (Traefik)
Run Stacks (CrowdSec)
Run Applications (AppSec)

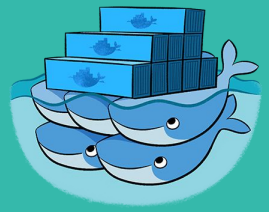

Users / Internet
openSUSE
Docker Swarm Cluster
Allowed HTTPS Traffic
HTTPS
Restricted / Malicious Traffic
HTTPS
Traefik Stack
træfik
Blocked by CrowdSec
CrowdSec Stack
CrowdSec
cscli decisions list
IP ADDRESS TYPE DURATION REASON
10.10.10.21 ban 4h http-bad-user-agent
203.0.113.45 captcha 1h crowdsecurity/nginx-bf
198.51.100.8 ban 24h crowdsecurity/http-probing
Running Apps (Protected by AppSec)
AppSec
Blocked by AppSec
Docker Swarm Nodes (openSUSE)
Manager Node 1
openSUSE
docker
Swarm Manager
Scheduling
Cluster Management
Worker Node 1
openSUSE
docker
Run Stacks (Traefik)
Run Stacks (CrowdSec)
Run Applications (AppSec)
Worker Node 2..N
openSUSE
docker
Run Stacks (Traefik)
Run Stacks (CrowdSec)
Run Applications (AppSec)

Topics

- openSUSE based Docker Swarm (*or K8S*) Cluster
- Deploy & Setup Traefik
- Integrate CrowdSec into Traefik
- Deploy Apps
- Secure Apps
- Demo!

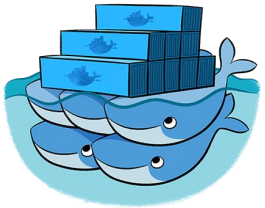
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openSUSE based Docker Swarm (*or K8S*) Cluster





Docker Swarm Cluster



HOSTNAME	STATUS	AVAILABILITY	MANAGER STATUS	ENGINE VERSION
tw-00	Ready	Active	Leader	29.4.0-ce
tw-01	Ready	Active	Reachable	29.4.0-ce
tw-02	Ready	Active	Reachable	29.4.0-ce

K8S Cluster



NAME	STATUS	ROLES	AGE	VERSION	OS-IMAGE
m01.rke2	Ready	control-plane,etcd,master	431d	v1.35.6+rke2r1	openSUSE MicroOS
m02.rke2	Ready	control-plane,etcd,master	431d	v1.35.6+rke2r1	openSUSE MicroOS
m03.rke2	Ready	control-plane,etcd,master	431d	v1.35.6+rke2r1	openSUSE MicroOS

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Deploy and Setup Traefik & Integrate CrowdSec into Traefik





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Prepare the PATHs



```
$ sudo mkdir -p /etc/traefik/coraza/rules
$ sudo mkdir -p /etc/traefik/configurations
$ sudo mkdir /var/log/traefik
```

CrowdSec

```
$ sudo mkdir -p /etc/crowdsec/secrets
$ sudo mkdir -p /var/lib/crowdsec/data
```

Prepare the PATHs



```
/etc/traefik/
├── acme.json
├── configurations
│   ├── coraza-waf.yml
│   ├── your-apps-com.yml
│   └── middlewares.yml
├── coraza
│   └── rules
└── traefik.yml
```

/etc/systemd/system/traefik.service

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```
[Unit]
Description=Traefik Gateway Container
After=network-online.target crowdsec.service
Wants=network-online.target crowdsec.service

[Service]
Restart=always
RestartSec=5

ExecStart=/usr/bin/podman container run \
  --name traefik \
  --replace \
  --network host \
  -v /etc/traefik/traefik.yml:/traefik.yml:Z \
  -v /etc/traefik/acme.json:/etc/traefik/acme.json:Z \
  -v /etc/traefik/configurations:/configurations:Z \
  -v /etc/traefik/coraza/rules:/etc/coraza/rules:ro,Z \
  -v /var/log/traefik:/logs:Z \
  docker.io/library/traefik:v3.7 \
  --configFile=/traefik.yml

ExecStop=/usr/bin/podman container stop traefik

[Install]
WantedBy=multi-user.target
```



/etc/systemd/system/crowdsec.service



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```
[Unit]
Description=CrowdSec Security Engine Container
After=network-online.target
Wants=network-online.target

[Service]
Restart=always
RestartSec=5

[Service]
ExecStart=/usr/bin/podman container run \
  --name crowdsec \
  --replace \
  --network host \
  -v /var/log/traefik:/var/log/traefik:ro,z \
  -v /var/lib/crowdsec/data:/var/lib/crowdsec/data:Z \
  -v /etc/crowdsec:/etc/crowdsec:Z \
  -v /etc/crowdsec/acquis.yaml:/etc/crowdsec/acquis.yaml:ro,z \
  -e COLLECTIONS="crowdsecurity/traefik crowdsecurity/appsec-generic-rules" \
  -e APPSEC_ENABLED=true \
  -e FEATURE_FLAGS=appsec \
  docker.io/crowdsecurity/crowdsec:latest

ExecStop=/usr/bin/podman container stop crowdsec

[Install]
WantedBy=multi-user.target
```



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Traefik Middlewares

/etc/traefik/configurations/middlewares.yml



https://gitlab.com/andisugandi/diy-lb/-/blob/main/traefik/dynamic/middleware.yml

CrowdSec Acquis Conf.

/etc/crowdsec/acquis.yaml




```
filenames:
  - /var/log/traefik/access.log
labels:
  type: traefik
```

Logrotate Traefik Logs

/etc/logrotate.d/traefik



```
/var/log/traefik/*.log {
    daily
    rotate 14
    compress
    missingok
    notifempty
    copytruncate
}
```

Start the Services

```
$ sudo systemctl enable --now traefik
```



```
$ sudo systemctl enable --now crowdsecc
```



CrowdSec

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Add CrowdSec Boucer

```
$ sudo podman exec crowdsec cscli \
    bouncers add traefik-bouncer
```

Output:

```
API key for 'traefik-bouncer':

***-it-is-a-secret-key-keep-it-private-***

Please keep this key since you will not be able to retrieve it!
```

Traefik Middlewares (Update!)[1]

/etc/traefik/configurations/middlewares.yml

```
http:
  middlewares:
    crowdsec-bouncer:
      plugin:
        crowdsec-bouncer:
          enabled: true
          useRealIP: true
          crowdsecLapiHost: 127.0.0.1:8081
          crowdsecLapiKey: {{ change-it-with-your-own-lapi-key }}
```





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Deploy Apps



kubectl describe ingressroute -n your-apps-com


```
Name:         your-apps-com-ingressroute
Namespace:    your-apps-com
Labels:       <none>
Annotations:  <none>
API Version:  traefik.io/v1alpha1
Kind:         IngressRoute
Metadata:
  Creation Timestamp:  2026-08-04T15:24:42Z
  Generation:          2
  Resource Version:    57507780
  UID:                 2c061e8f-e059-4d7e-85c2-ac199875fb4d
Spec:
  Entry Points:
    web
  Routes:
    Kind:   Rule
    Match:  Host(`your.apps.com`)
    Middlewares:
      Name:  kh-basic-auth
    Services:
      Name:  your-apps-com-frontend
      Port:  80
Events:       <none>
```

/etc/traefik/configurations/your.apps.com.yml


```
http:
  routers:
    longhorn-ui-router:
      entryPoints:
        - websecure
      rule: Host(`your.apps.com`)
      service: your-apps-com-backend
      middlewares:
              - chain-frame-sameorigin@file
      tls:
        certResolver: letsencrypt

  services:
    your-apps-com-backend:
      loadBalancer:
        servers:
          - url: "http://10.0.0.1:80"
```

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Secure Apps

Traefik Middlewares (Update!)[2]

/etc/traefik/configurations/middlewares.yml

```
http:
  middlewares:
    crowdsec-appsec:
      plugin:
        crowdsec-bouncer:
          enabled: true
          useRealIP: true
          crowdsecLapiHost: 127.0.0.1:8081
          crowdsecLapiKey: {{ change-it-with-your-own-lapi-key }}
          crowdsecAppsecEnabled: true
          crowdsecAppsecHost: 127.0.0.1:7422
```

Traefik Middlewares (Update!)[3]

/etc/traefik/configurations/middlewares.yml

```
http:
  middlewares:
    chain-frame-sameorigin:
      chain:
        middlewares:
          - limit-bots
          - headers-frame-sameorigin
          - gacor-regex
          - crowdsec-bouncer
          - crowdsec-appsec
```

CrowdSec Acquis Conf. (Update!)

/etc/crowdsec/acquis.yaml

```
filenames:
  - /var/log/traefik/access.log
labels:
  type: traefik
---
source: appsec
listen_addr: 127.0.0.1:7422
appsec_config: crowdsecurity/appsec-default
labels:
  type: appsec
```

cscli commands




```
$ sudo podman exec crowdsec \
    cscli collections install crowdsecurity/appsec-virtual-patching

$ sudo podman exec crowdsec cscli \
    collections install crowdsecurity/appsec-generic-rules
```

Demo!

Thank You!



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